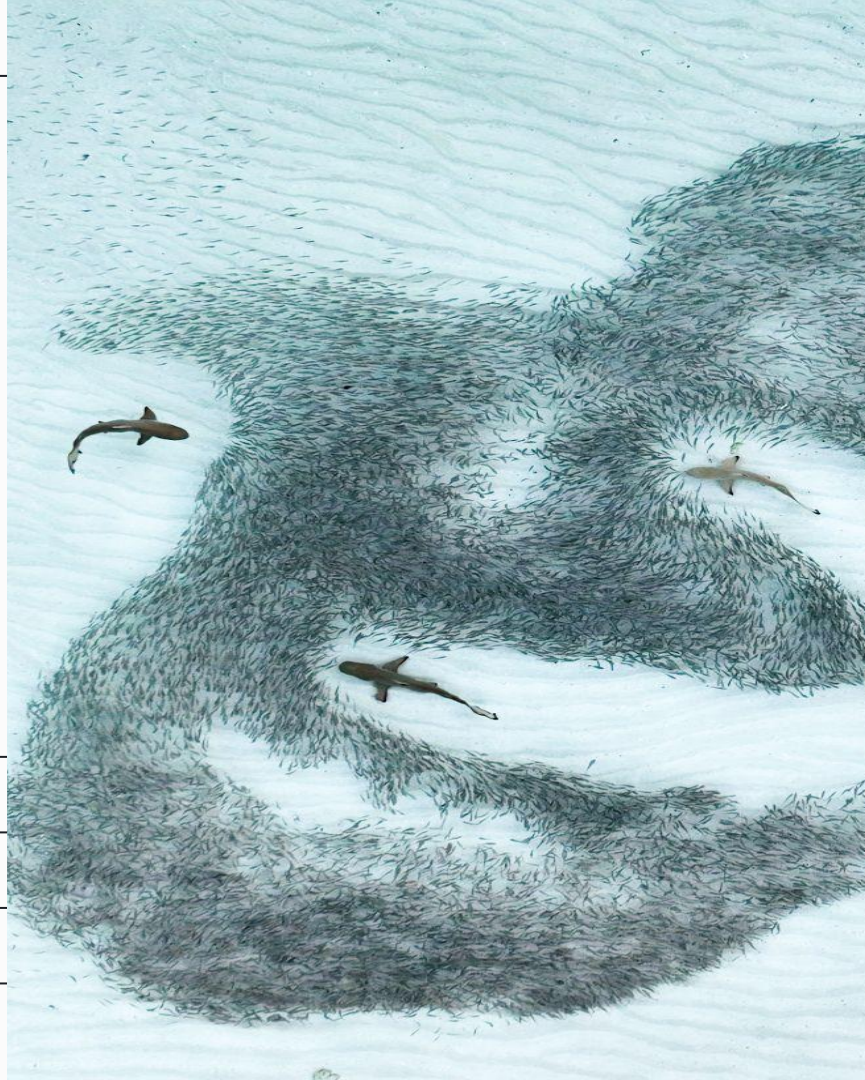


Understanding Predator Prey Interactions from Videos

Nicole Kloss

Second Thesis Update

August 2026



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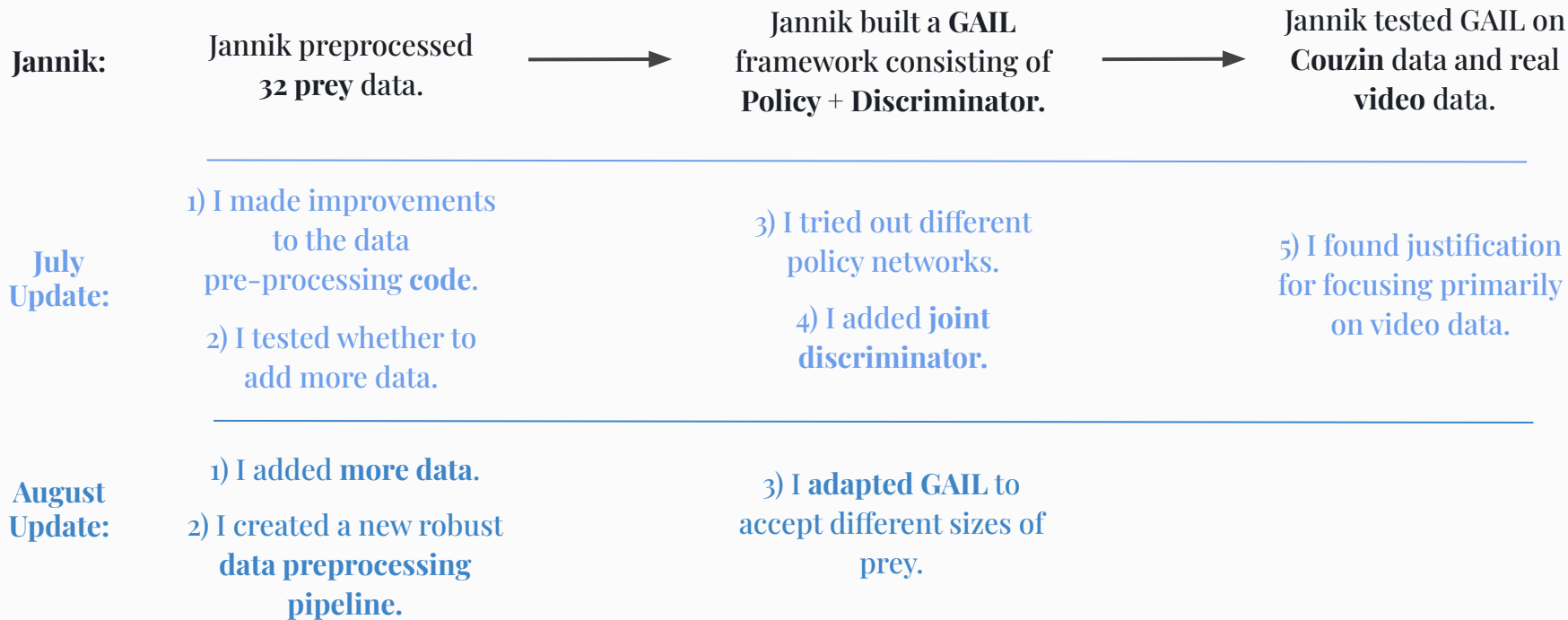
05

Next Steps

Recap

As a reminder, Jannik managed to build an imitation framework that captured individual movement.

My thesis extends his work aiming to capture the interaction between predator and prey.



Data Processing

Focus : 16 + 32 prey condition

32 prey condition: 32 attacks + 35 interactions → 793 windows

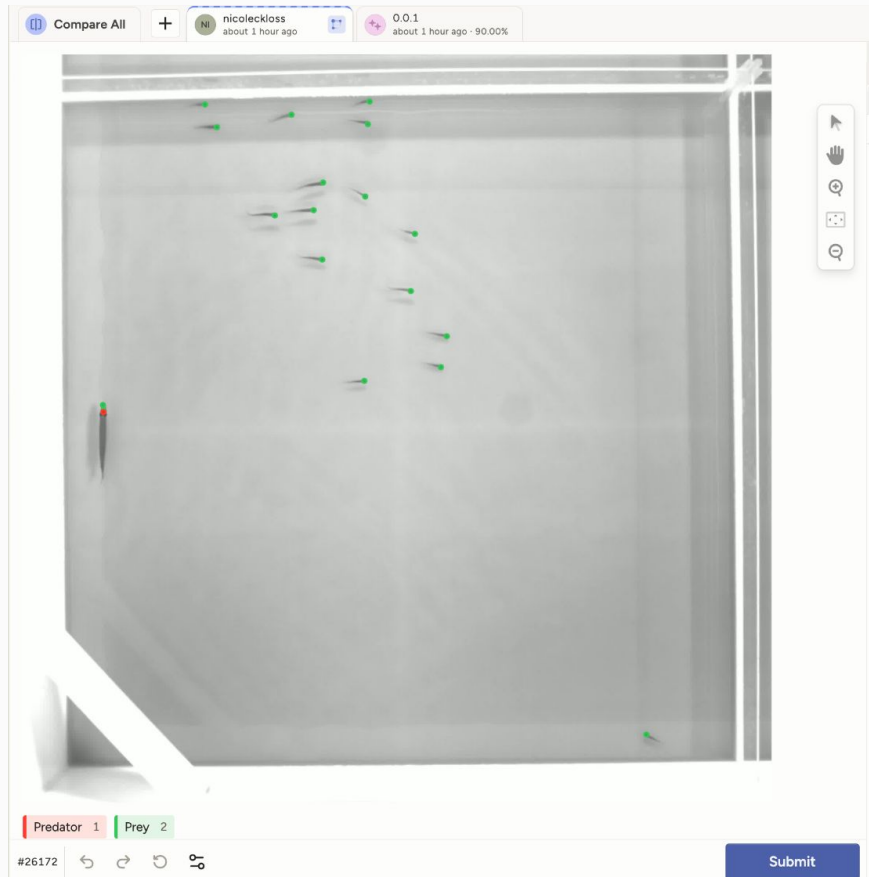
16 prey condition: 27 attacks + 40 interactions → 2,200 windows (so far)



Data Processing

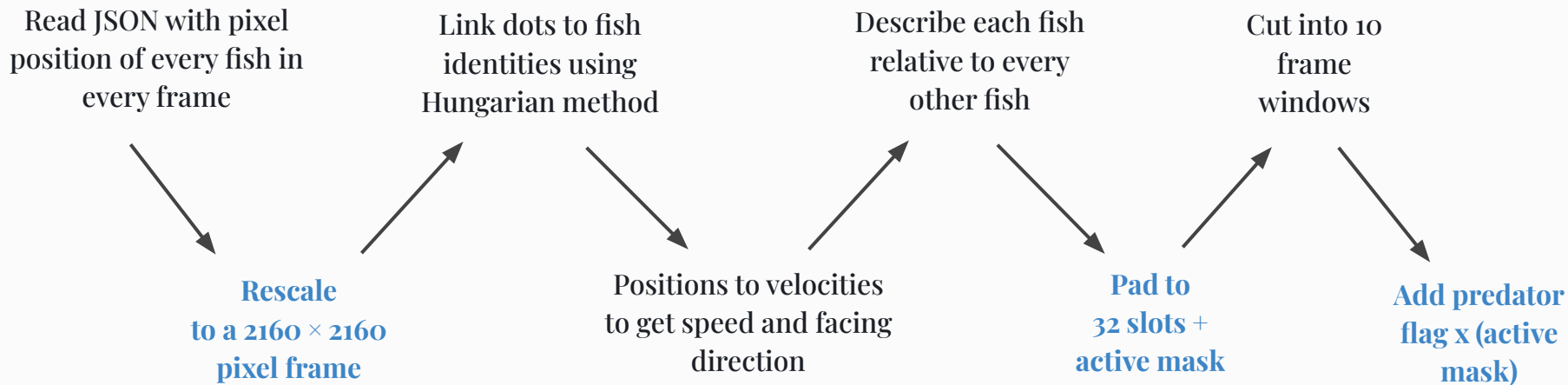
Hand-Labeled Both Attacks + Interactions

- I manually marked predator and prey positions frame by frame in Label Studio
- After ~300 labeled frames, I trained a YOLO model to aid with my labeling
- Resulted in majority of frame being pre-labeled
- Each labeled attack/interaction is exported into a JSON file



Data Processing

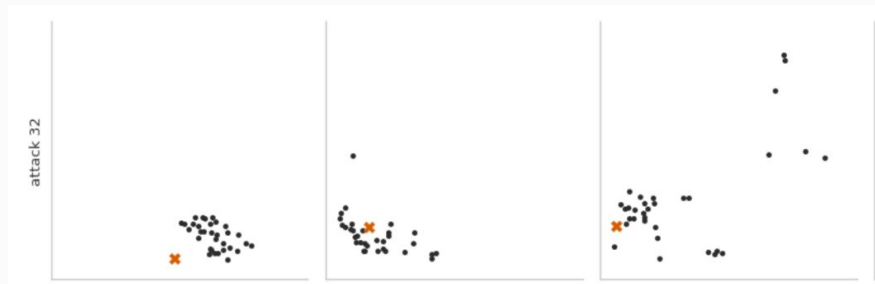
Generated Pred Prey Tensors



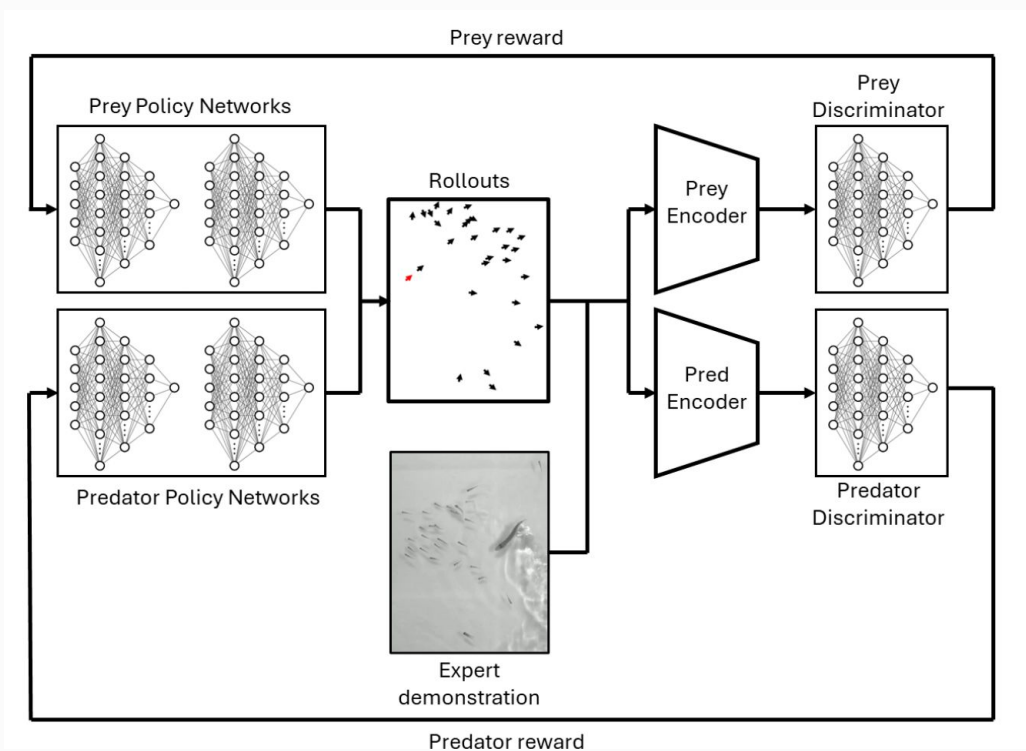
Data Processing

Created Initial Position Pool

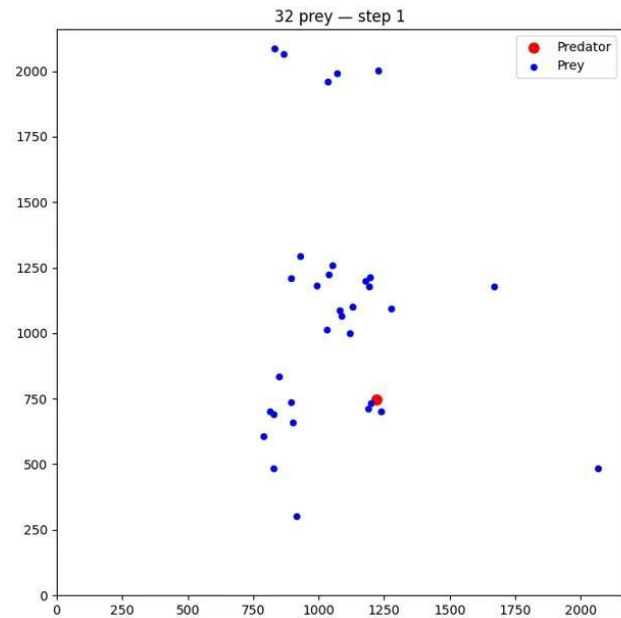
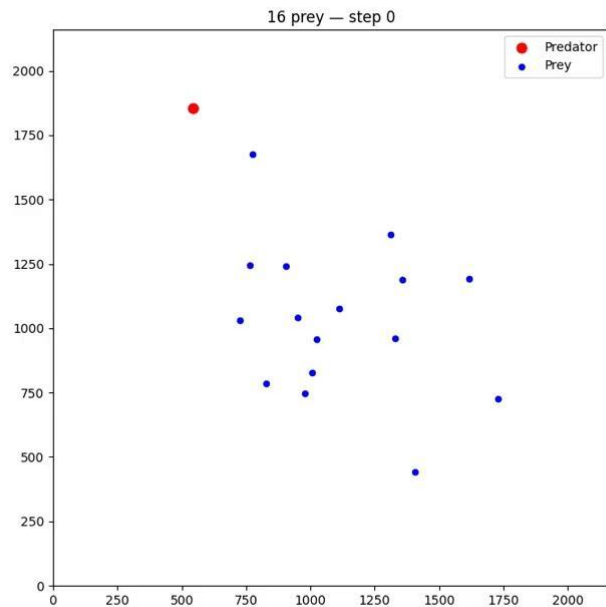
- Generates a bank of real starting positions (fish positions + headings) taken from my data
- Allows the policy and discriminator to start from realistic configurations rather than random positions



Rebuilding GAIL

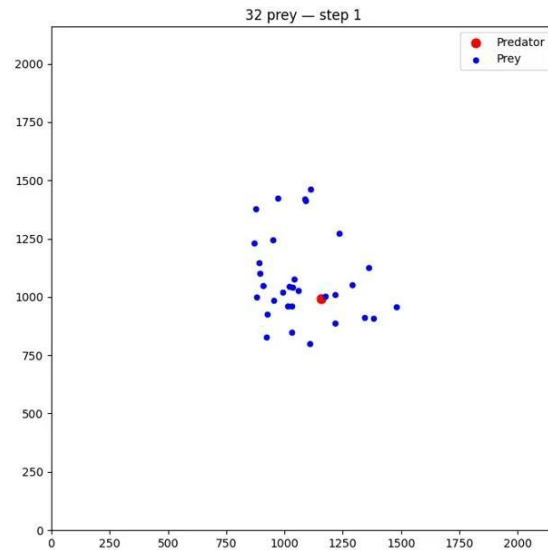
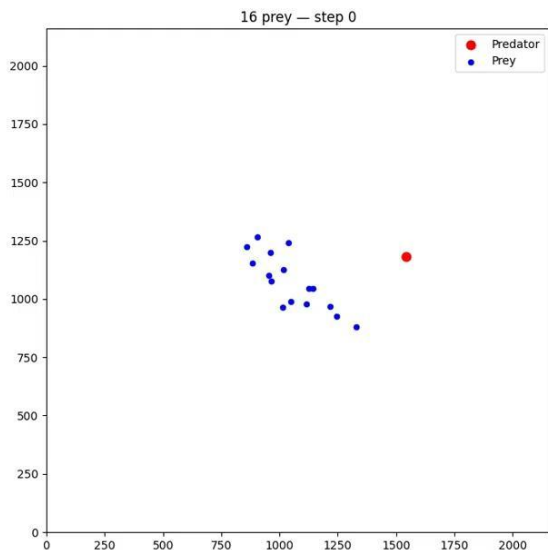


BC on 16+32 Data

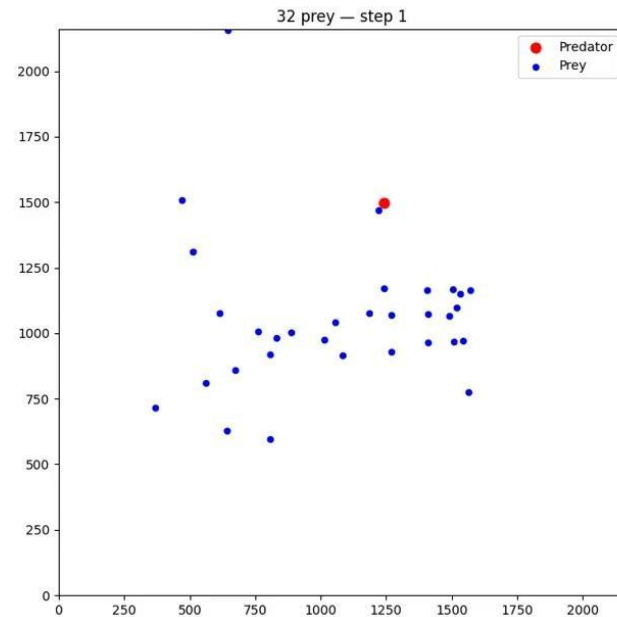
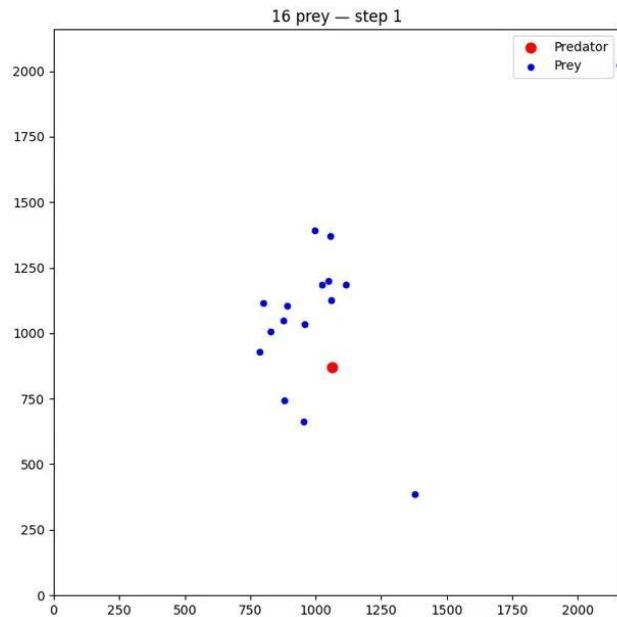


Prey Speed Distribution: Interaction vs Attack Data

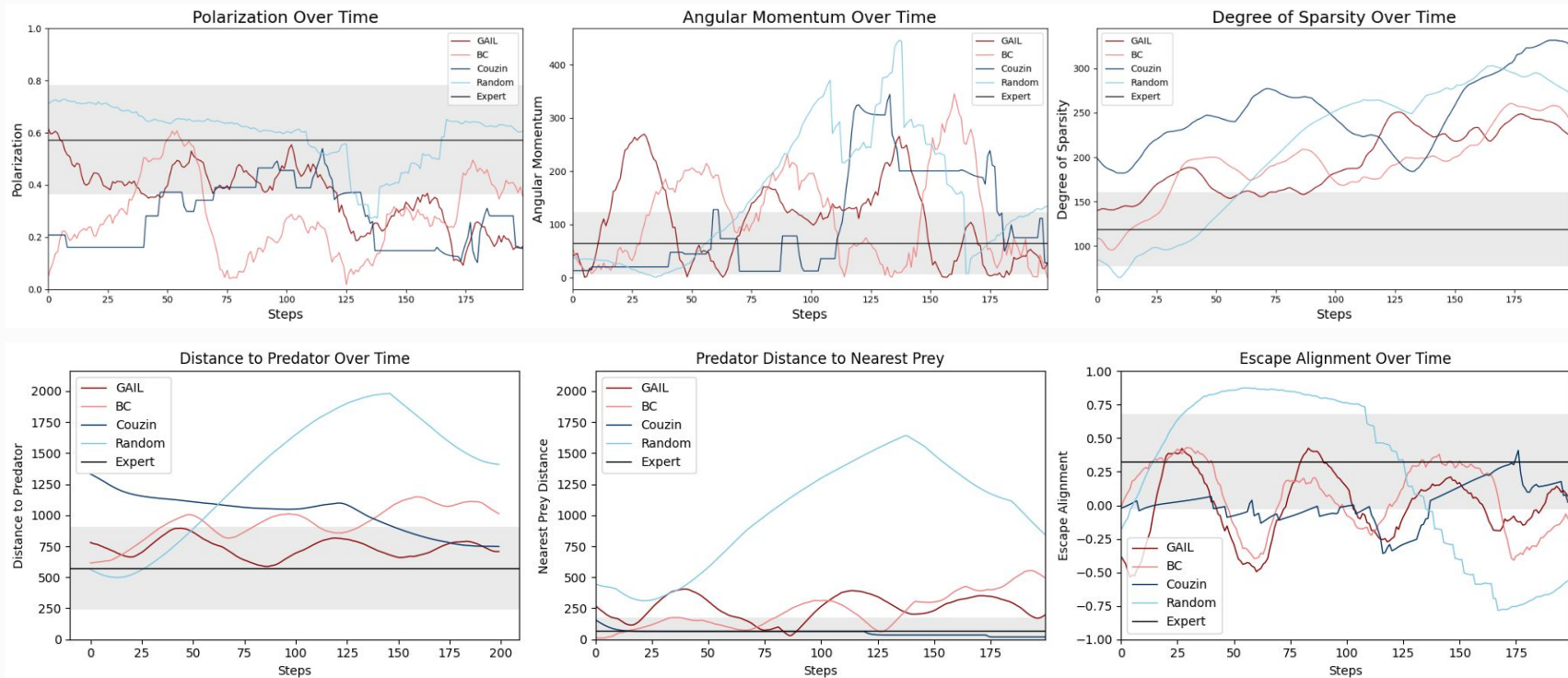
interaction	16 prey	769 win	median= 1.68	p95= 3.73	per-window peak median= 3.88	max= 13.70
interaction	32 prey	24 win	median= 1.72	p95= 3.17	per-window peak median= 3.52	max= 4.84
attack	16 prey	1431 win	median= 3.59	p95=14.47	per-window peak median=19.41	max= 35.36
attack	32 prey	769 win	median= 5.32	p95=17.58	per-window peak median=31.77	max= 35.36



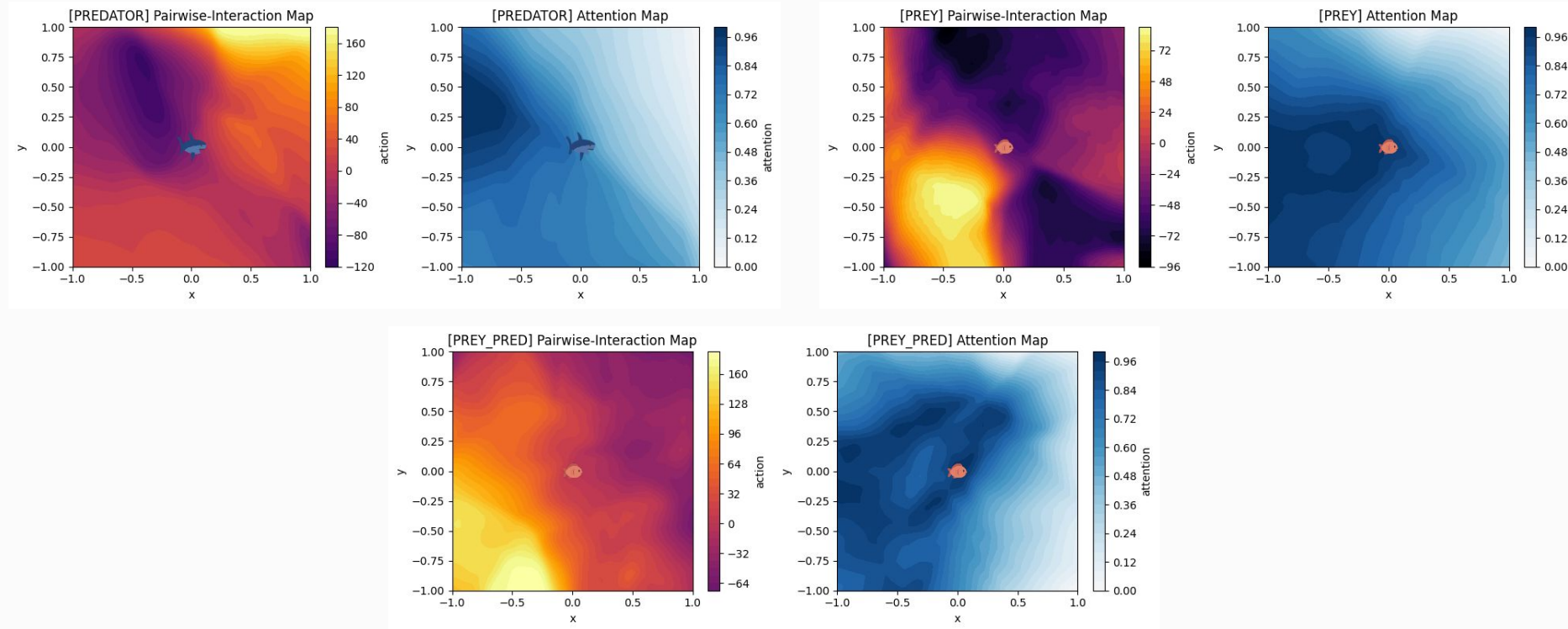
GAIL on 32-16 model [visualization]



GAIL on 32-16 model [swarm metrics]



GAIL on 32-16 model [network analysis]



Next steps

1. Better define a successful GAIL model

2. Reincorporate the joint discriminator + Couzin data

3. Complete the interaction hand-labeling

Explanatory Slides from July Presentation:

GAIL

We have expert demonstrations, but no reward function, and we want a policy that reproduces the expert's behavior. This is an imitation learning problem, so the question is: which method should we use?

Behavior Cloning? **Simple but requires a lot of demonstration data and errors can compound pushing the agent into unseen states.**

Inverse Reinforcement Learning? **Tries to infer the reward but many rewards can explain the same demonstrations, resulting in ambiguity and high computational complexity.**

Generative Adversarial Imitation Learning? **Avoids defining/recovering explicit reward and instead learns a policy by matching expert-like trajectories through adversarial imitation. This avoids compounding errors and is substantially faster than IRL.**

